Causal Characterization of Measurement and Mechanistic Anomalies

Author information scrubbed for double-blind reviewing

No Institute Given

1 Introduction

Real-world datasets often contain anomalies, samples generated by processes that deviate from the norm. Such deviations may reflect measurement or data-collection errors, such as faulty sensors or misrecorded values, or genuine changes in the underlying data-generating mechanism. We call these two cases *measurement anomalies* and *mechanistic anomalies*. Distinguishing them is practically important. An anomalous pressure reading may call either for sensor recalibration or for intervention in the physical process itself.

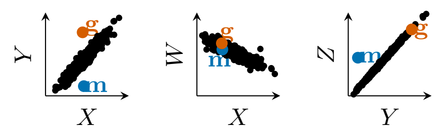


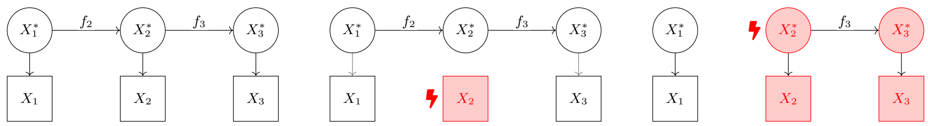
Fig. 1: Different outlier types under the causal DAG $W \leftarrow X \rightarrow Y \rightarrow Z$.

We give an example in Fig. 1. In the left plot, points **g** and **m** stand out, but it is unclear whether they reflect measurement error or a mechanism shift. Examining additional *causal* relationships clarifies the difference. Neither point is anomalous for $X \rightarrow W$, but **m** violates $X \rightarrow Y$ and $Y \rightarrow Z$ in a way that we can 'normalize' it for both simply by adjusting the value for Y, which suggests Y was measured wrongly. **g** on the other hand, behaves normally for $X \rightarrow W$ and $Y \rightarrow Z$, suggesting the mechanism of $X \rightarrow Y$ was changed.

Explainable anomaly detection typically attributes an anomaly score to input features [10,4], whereas causal root cause analysis seeks variables that caused the deviation in the data-generating process [1,3,6,5]. Existing RCA methods, however, do not explicitly distinguish measurement failures from mechanistic changes. This distinction matters because the same observed outlying value can imply different interventions depending on whether the latent causal state or only its observed readout was perturbed. We bridge this gap by modeling mechanistic and measurement anomalies as interventions on different levels of a causal graph $\mathcal{G}$: mechanistic anomalies intervene on latent "real" variables, while measurement anomalies intervene on observed "measured" variables.

2 Causal Model and Inference

We first introduce our causal model for mechanistic and measurement anomalies, and then discuss identifiability and inference under this model.



(a) Clean process (b) Measurement Outlier (c) Mechanistic Outlier

Fig. 2: Different outlier types induce different downstream behavior. ⚡ indicate interventions that correspond to measurement outliers (intervention on observed variables) and mechanistic outliers (intervention on latent variables).

Causal model Let $\mathbf{X}^* = (X_1^*, \dots, X_d^*)$ denote latent causal state variables and $\mathbf{X} = (X_1, \dots, X_d)$ their observed readouts. Most samples are generated by an anomaly-free process, while a small fraction contain sparse deviations that we call anomalies. Let $\mathbf{V} = \mathbf{X}^* \cup \mathbf{X}$ and let $\mathcal{G} = (\mathbf{V}, E)$ be a DAG. The latent variables $\mathbf{X}^*$ form the causal system, and each observation X_j is a readout of exactly one latent variable X_j^*; see Fig. 2a. We write Pa_j^* for the latent parents of X_j^* and assume causal sufficiency and faithfulness conditions over the latent causal graph $\mathcal{G}[\mathbf{X}^*]$ [7]. In the anomaly-free regime, the data are generated by a structural causal model with independent noises, $X_j^* = f_j(\mathrm{Pa}_j^*, U_j)$, and measurements are faithful, $X_j = X_j^*$. The clean joint distribution factorizes as $P(\mathbf{X}^*, \mathbf{X}) = \prod_{j=1}^{d} P(X_j^* \mid \mathrm{Pa}_j^*)\, P(X_j \mid X_j^*)$. Anomalies are modeled as sample-level violations of this clean process.

Definition 1 (Mechanistic Anomaly). *A mechanistic anomaly at variable j replaces the clean latent mechanism by an exogenous outlier draw $O_j^{mech} \sim P_{O_j^{mech}}$. With latent indicator $Z_j \sim \mathrm{Bernoulli}(\pi_j^{mech})$, $X_j^* = (1 - Z_j) f_j(\mathrm{Pa}_j^*, U_j) + Z_j O_j^{mech}$. Thus, when $Z_j = 1$, the latent causal state itself is perturbed and the effect may propagate to descendants.*

Definition 2 (Measurement Anomaly). *A measurement anomaly at variable j replaces only the observed readout by an exogenous outlier draw $O_j^{meas} \sim P_{O_j^{meas}}$. With latent indicator $W_j \sim \mathrm{Bernoulli}(\pi_j^{meas})$, $X_j = (1 - W_j) X_j^* + W_j O_j^{meas}$. Thus, when $W_j = 1$, the latent causal state remains unchanged and the perturbation does not propagate.*

The variables O_j^{mech} and O_j^{meas} are independent of the clean SCM variables and noises. We interpret both anomaly types as hard interventions [7]. A mechanistic anomaly cuts the incoming causal edges into X_j^* and replaces its structural assignment, whereas a measurement anomaly cuts only the measurement edge $X_j^* \to X_j$. This distinction is central. Since observed variables have no children, a measurement anomaly affects only the coordinate X_j. By contrast, a mechanistic anomaly changes the latent state X_j^* and may therefore alter all downstream variables in the latent causal graph (see Fig. 2).

Finally, anomalies are assumed to be rare and sparse. Formally, the clean assignment occurs for most samples, $P((\mathbf{W}, \mathbf{Z}) = \mathbf{0}) \gg 0.5$, and we instantiate the SMS hypothesis [12] by placing an independent Bernoulli prior over the anomaly indicators $(\mathbf{W}, \mathbf{Z})$. This prior favors explanations with few root causes unless the observed evidence strongly suggests otherwise.

Identifiability and Inference We expect anomalies to be rare, too rare to estimate separate replacement distributions for all failure types. We hence assume a fixed simple symmetric outlier distribution, $\tilde{P}_{O_j^{mech}} = \tilde{P}_{O_j^{meas}}$, to be specified by the user. Further, we assume an estimate of the anomaly-free SCM density, obtained from the majority of anomaly-free data.

First, we have an identifiability limit. If j is a sink in $G[X^*]$, then a mechanistic anomaly at X_j^* and a measurement anomaly at X_j have the same observable effect. Neither can propagate to descendants, and both replace only the observed coordinate X_j. Hence at sinks the two are observationally indistinguishable.

Second, we have a positive identifiability result. The latent intervention probabilities are generically distinguishable at non-sink nodes through their different downstream effects. We show this for linear, and polynomial additive-noise SCMs with Gaussian clean noise and nodewise identical Gaussian replacement distributions up to a Lebesgue-null set of model parameters.

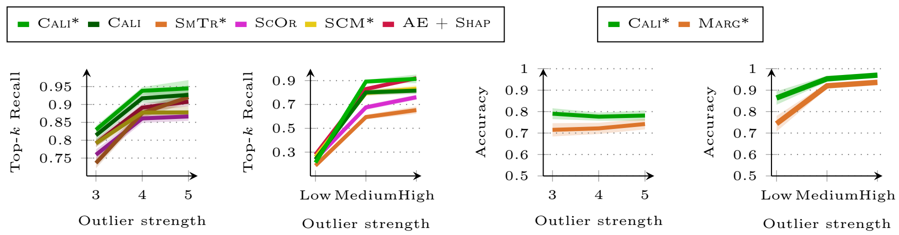
Even when identifiable, exact Bayesian inference over anomaly assignments is computationally expensive. The number of assignments grows exponentially in the number of variables, and measurement anomalies require marginalizing over unobserved pre-replacement values. We therefore use a hard-assignment approximation, $\log \sum_a \mathcal{L}(x; a, \pi) \approx \max_a \log \mathcal{L}(x; a, \pi)$, which replaces the posterior mixture by the most likely sparse explanation. For fixed assignments, the optimal intervention probabilities are the empirical frequencies, so our algorithm Cali profiles them out and obtains an adaptive sparsity penalty over candidate explanations. Empirically, it matches the model-optimal classification rule with oracle $\boldsymbol{\pi}$ where exact inference is feasible, while scaling to real-world problems.

3 Empirical Results

We evaluate Cali on synthetic SCMs with known ground truth, real-world causal benchmarks, and a case study. We test two tasks, root-cause localization and anomaly-type classification.

Quantitative results. We compare Cali to the RCA baselines SCM* [1,13], SmTr, ScOr [6], and AE-Shap (autoencoder with Shapley values). We give the results in Fig. 3. For localization, Cali performs best on synthetic data and remains competitive with existing methods on real-world benchmarks. For classification, i.e. telling whether it was an measurement or mechanistic anomaly, Cali consistently outperforms the baseline Marg*, which classifies by checking propagation of marginal outlierness.

Case study In Tab. 1, we show high-confidence anomalies found by Cali in a subsample of the NYC Yellow Taxi dataset [2]. The detected measurement



(a) RCA: synthetic (b) RCA: real (c) Type: synthetic (d) Type: real

Fig. 3: Quantitative evaluation. (a,b) Root-cause localization (Top-k recall, where k is the true number of root causes). (c,d) Anomaly type classification. We measure whether localized root causes are correctly labeled as measurement or mechanistic anomalies. Asterisks denote access to the ground-truth DAG; other methods are non-causal or use a learned DAG.

Table 1: Measurement (left) and mechanistic (right) anomalies identified by Cali in the NYC Taxi dataset. The detected root cause is bolded. Coordinates are summarized by Air Distance.

Distance	Air D.	Time	Fare	Tip	Total
3.90	3.16	**94.65**	10.00	2.00	13.30
9.40	0.01	0.20	2.50	0.00	3.30
0.60	0.98	**33.52**	5.00	0.00	6.30

Distance	Air D.	Time	Fare	Tip	Total
0.00	0.06	0.75	**220.00**	44.05	264.35
0.00	0.00	0.30	**52.00**	0.00	58.13
1.80	1.39	8.48	**52.00**	11.63	69.76

anomalies are local inconsistencies in recorded attributes: for example, a trip with recorded distance 9.4 miles but air distance 0.01 miles, duration 0.2 minutes, and minimum fare is best explained as a distance-recording error. The detected mechanistic anomalies instead indicate changes in the fare-generating process. For example, the fares of exactly \$52 are consistent with the JFK flat-fare rule, where fare is set by a special pricing mechanism rather than the usual distance–time relationship. Thus, Cali separates likely data-entry or sensor errors from genuine changes in the underlying process.

4 Conclusion

We studied root cause analysis from a new perspective, distinguishing measurement errors from genuine mechanistic anomalies by modeling both as interventions on latent and observed variables. We achieve competitive localization performance, enable the novel task of outlier type classification, and demonstrated in a case study that our approach produces plausible and actionable anomaly characterizations in real-world data.

References

1. Budhathoki, K., Minorics, L., Blöbaum, P., Janzing, D.: Causal structure-based root cause analysis of outliers. In: International conference on machine learning. pp. 2357–2369. PMLR (2022)
2. Elemento: NYC Yellow Taxi Trip Data. `https://www.kaggle.com/datasets/elemento/nyc-yellow-taxi-trip-data` (nd), kaggle dataset (accessed 2025-11-11).
3. Han, X., Zhang, L., Wu, Y., Yuan, S.: On Root Cause Localization and Anomaly Mitigation through Causal Inference. In: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management. p. 699–708. CIKM '23, Association for Computing Machinery, New York, NY, USA (2023)
4. Lundberg, S.M., Lee, S.I.: A unified approach to interpreting model predictions. In: Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS) (2017)
5. Nguyen, P., Tran, T., Gupta, S., Nguyen, T., Venkatesh, S.: Root Cause Explanation of Outliers under Noisy Mechanisms. Proceedings of the AAAI Conference on Artificial Intelligence **38**, 20508–20515 (2024)
6. Orchard, W.R., Okati, N., Mejia, S.H.G., Blöbaum, P., Janzing, D.: Root Cause Analysis of Outliers with Missing Structural Knowledge. In: The Thirty-ninth Annual Conference on Neural Information Processing Systems (2025)
7. Pearl, J.: Causality: Models, Reasoning and Inference. Cambridge University Press, 2nd edn. (2009)
8. Perry, R., Von Kügelgen, J., Schölkopf, B.: Causal discovery in heterogeneous environments under the sparse mechanism shift hypothesis. Advances in Neural Information Processing Systems **35**, 10904–10917 (2022)
9. Peters, J., Janzing, D., Schölkopf, B.: Elements of causal inference: foundations and learning algorithms. The MIT press (2017)
10. Ribeiro, M.T., Singh, S., Guestrin, C.: Why Should I Trust You?: Explaining the Predictions of Any Classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. p. 1135–1144. KDD '16, Association for Computing Machinery, New York, NY, USA (2016)
11. Schölkopf, B., Janzing, D., Peters, J., Sgouritsa, E., Zhang, K., Mooij, J.: On causal and anticausal learning. arXiv preprint arXiv:1206.6471 (2012)
12. Schölkopf, B., Locatello, F., Bauer, S., Ke, N.R., Kalchbrenner, N., Goyal, A., Bengio, Y.: Towards Causal Representation Learning. ArXiv **abs/2102.11107** (2021)
13. Strobl, E., Lasko, T.A.: Sample-specific root causal inference with latent variables. In: Conference on Causal Learning and Reasoning. pp. 895–915. PMLR (2023)